

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures

COURSE CODE: CSA03

Register Number	192525171
Name	TEJASWINI.P

COURSE OUTCOME COVERED

CO1: *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)*

Assessment Tool Weightage: 35%

Questions:2 EACH QUESTIONS CARRIES: 10 MARKS Total Marks:20 Duration: ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

I Tejaswini(192525171) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Tejaswini

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

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ANS: .

Data → Data Object: represents meaningful information

Data Object → Data Structure: organizes related data

Data Structure → ADT: provides a foundation for implementing abstract operations

ADT → Operations: defines permitted operations

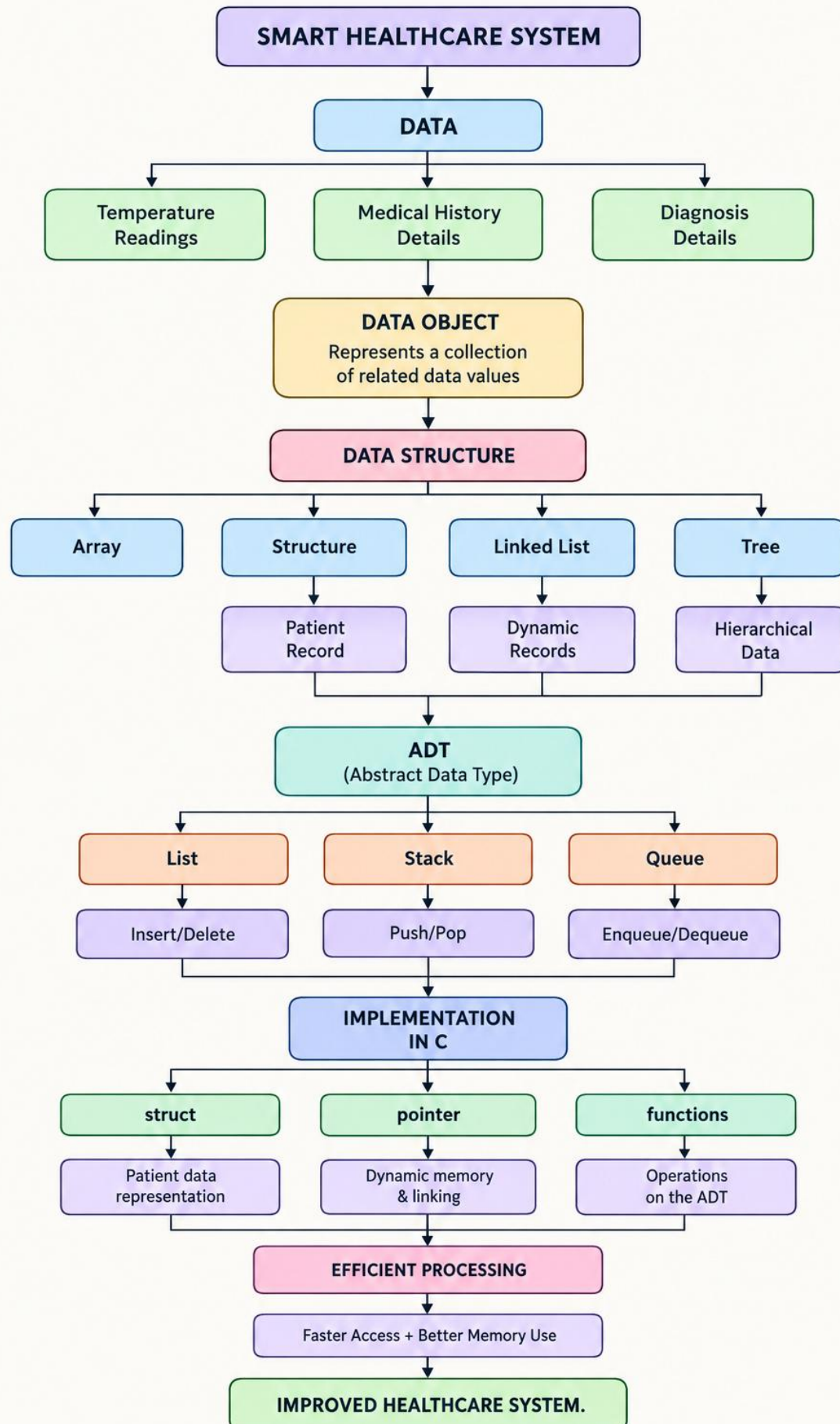
ADT → C Implementation: ADT operations are implemented using C

Pointers → Dynamic Memory: enable dynamic allocation and linking

Functions → Operations: perform insertion, deletion, searching, etc.

Efficient Processing → Improved System: reduces processing time and improves memory utilization

Concept	Healthcare Example
Data	Temperature = 38°C
Data Object	Patient record
Structure	struct Patient
Array	Store temperature readings
Linked List	Dynamic patient records
Stack	Recently accessed records
Queue	Patients waiting for consultation
Tree	Medical classification
Pointer	Links dynamically allocated records
ADT	List/Stack/Queue operations



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Data Structure	Traffic Application
Integer	Number of vehicles
Float	Vehicle speed
Character	Traffic signal status
Array	Store vehicle counts at multiple signals
Linked List	Dynamically maintain vehicle records
Queue	Vehicles waiting at a signal
Stack	Recent traffic events/history
Tree	Hierarchical road/signal classification
Graph	Road network and intersections

Primitive data structures store basic single values such as integers, characters and floating-point numbers.

Non-primitive data structures organize multiple values and provide more complex ways of managing data.

Linear structures arrange elements sequentially. Each element generally has a predecessor and successor, except the first and last elements. Examples include **arrays, linked lists, stacks and queues**.

Non-linear structures do not store elements in a simple sequential order. They represent hierarchical or network relationships. **Trees** represent hierarchy, while **graphs** represent connections between entities.

Efficiency

- **Arrays** provide fast direct access using an index.
- **Linked lists** allow flexible dynamic insertion and deletion.
- **Queues** are suitable for vehicles waiting at traffic signals because they follow **FIFO**.
- **Trees** efficiently represent hierarchical information.
- **Graphs** are ideal for road networks because roads can be represented as **edges** and intersections as **vertices**.
- Appropriate selection of a structure improves **time efficiency, memory utilization and traffic-management performance**.

